Custom Encryption Algorithm

Project Documentation

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Abstract

This report seeks to research and develop a new algorithm based on an already established cipher. As there are many ciphers today, almost all of them have some weaknesses. They require improvements to make them more secure. This report introduces a new algorithm based on Atbash Cipher. Atbash Cipher is one of the oldest ciphers that can easily be broken today. It is simple and easily decipherable which is why the new algorithm has new mathematical and logical operations included. There has been use of bitwise rotation and bitwise NOT operation as the new logical operations and calculations with key decimals, modifications on the original atbash formula and conversions between decimal and binary as the mathematical operations.

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1 Introduction

1.1 Overview on Security

Security in general means to give protection to something valuable from danger and harm. While there are many types of Securities in this world, one that has grown with the increase of technologies and innovations is information security. Information security is the practice of protecting information from unauthorised access, unauthorised alterations, malicious acquisition and unavailability. This type of security ensures that only the authorised or rightful people can access the information (Holdsworth & Kosinski, 2025).

1.2 CIA triad

The main purpose of Information security is to maintain the CIA triad. The CIA triad includes Confidentiality, Integrity and Availability which collectively represent the fundamental security objectives for data and information protection. Confidentiality ensures the information is not disclosed to unauthorised persons and assures the power of the information holder. Integrity maintains the reliability of the information by making sure only authorised persons can make changes. And Availability makes sure that the information is available when it is needed (Stallings, 2020).

If the CIA triad is protected properly, the users can guarantee quality information. One of the most crucial parts in maintaining this CIA triad is Cryptography. Cryptography is the art or the science of encrypting a plain readable message to an unintelligible message and then decrypting it to its original form to be readable again (Rao, et al., 2025).

1.3 Aim and Objectives

1.3.1 Aim

The main aim of this report is to research and develop a new cryptographic algorithm based on an already established one.

1.3.2 Objectives

- Study the evolution of ancient cryptography to modern cryptography
- Comprehend the concepts of symmetric and asymmetric encryption systems
- Explore about Atbash cipher and its advantages and disadvantages
- Implement mathematical and logical operations on Atbash cipher

- Assess the implementations in the new algorithm
- Evaluate the new algorithm's advantages and disadvantages

1.4 Technical terminologies

Plaintext: Plaintext is the original intelligible message that is given as an input when encryption algorithm begins. It is the message that is meant to be seen by the receiver (Naser, 2021).

Key: Key is a unique piece of information that is used to encrypt information and decrypt the same information. Every key gives a (Naser, 2021) different type of ciphertext when used in encryption algorithm. The same key must be used to get the same plaintext (Naser, 2021).

Encryption: Encryption is the process of turning a plaintext into a ciphertext. It occurs at the sender side of the communication. It is used to hide the actual information from any third-party members (Naser, 2021).

Ciphertext: Ciphertext is an unintelligible and scrambled piece of message that is produced after the encryption process on a plaintext. It is the encrypted output of the plaintext. The ciphertext depend upon the plaintext and the key that was used in the encryption process (Naser, 2021).

Decryption: Decryption is the process of turning a ciphertext back into the plaintext. It occurs at the receiver side of the secret communication. It is used to reveal the actual information that was sent secretly by the sender. It is the reverse of encryption (Naser, 2021).

Cryptographic algorithm: Cryptographic algorithm is a set of procedures that may include logical and mathematical operations to encrypt and to decrypt the message. This set of procedures help make cryptography easier and more understandable (geeksforgeeks , 2025).

2 Introduction to Cryptographic systems

2.1 History of Cryptography

The history of cryptography began in the regions of Egypt, Greece and Rome for civilizations to communicate easily and safely. The word cryptography comes from the Greek words “kryptos” which means hidden and “graphien” means writing. This gives the definition of cryptography in simple terms, a hidden writing (Naser, 2021). The ability to send messages in secret from one place to the another has been a practice for thousands of years (Shores, 2020).

One of the earliest recorded forms of classical cryptography was by the Egyptians. They used the hieroglyphic substitution where some hieroglyphic letters were replaced with other symbols for secrecy. The secret message could only be deciphered if the receiver knows the cipher key (Ketha, 2024).

Figure 1: Hieroglyphics (Britannica, 2025)

The Egyptians occasionally used cryptographic hieroglyphics. This type of writing was found in the town of Menet on the tomb of an aristocrat, Khnumhotep II. The tomb was filled with the biography of the aristocrat which was made by his servants. These servants or copyists were the same people who were also in charge of writing cryptographic messages for their king during his reign. The cravings or pictures on the tomb were replaced with alternative symbols to hide the actual meaning, which today we refer to it as substitution ciphering (Pal, et al., 2020).

Figure 2: Hieroglyphics at tomb of Khnumhotep II (Kamrin, 2009)

Over at the Greek region, a new cryptography tool called the “scytale” was invented. A scytale is a cylinder or baton in Greek. The message would be written in a parchment strip wounded around the scytale and the strip would be sent by the sender without the scytale to the receiver. The receiver would have to find the same size of scytale and wound the strip to it to read or decipher the message (Louridas, 2025).

Figure 3: Scytale (rfelo, 2025)

The Spartans used this tool to keep secure communication with their military. Most people in ancient world weren’t as educated as today so, this method seemed to be very reliable for them (Pal, et al., 2020).

During the roman republican period, Julius Caesar made the contribution to cryptography by introducing the Caesar cipher. In this cipher, the letters in a word would be replaced by the letter after a predetermined number of shifts in the position. The fixed number is referred to as the key which is also used to decipher. This cipher was used by Julius Caesar to protect communication between him and his military and for diplomatic purposes as well (Ketha, 2024).

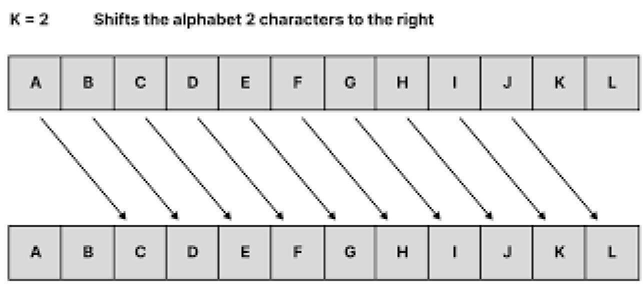

Figure 4: Caesar Cipher (vCollections, 2024)

*(See **Error! Reference source not found.** for Vigenère Cipher)*

The encryption methods kept getting stronger and sophisticated as the years passed by. Especially during the great wars, cryptography developed immensely as encryption was the crucial key to communicate with allies. The Germans created a machine called the Enigma Machine. It utilized rotors and complex wiring to perform substation and transposition on the letters of the message. It looked unbreakable due to the vast amount of possible cipher combinations it could do (Ketha, 2024). However, Alan Turing and his team developed the British Bombe, a machine that helped break Enigma codes. This is said to have shortened the war by 2 years (Fries, 2022).

Figure 5: Enigma Machine (Capstonex, 2022)

As it is known, the great wars didn't bring absolute peace in the world. The cold war made the USA to develop a very secure communication encryption standard called Data Encryption Standard (DES). It used a 56-bit key, which appeared to be cracked easily. That is why in 2000, it's much modified version Advanced Encryption Standard (AES) was introduced which supports keys up to 256 bits. This encryption standard is used worldwide due to its reliable security (Fries, 2022).

Finally, the modern ciphering techniques continue to become increasingly secure. There is new research in the areas concerning quantum and post-quantum cryptography algorithms which aim to bring greater security in the digital world (Sidhu, 2023).

2.2 Symmetric and Asymmetric Encryption

As the encryption systems advance, the importance of understanding symmetric and asymmetric encryption systems grow. The following are their explanations:

2.2.1 Symmetric encryption system

Symmetric encryption system is a type of encryption system that uses only one key for both encryption and decryption. It is one of the oldest forms of encryption systems like it was used in early substitution ciphers like the Caesar cipher. However, due to the simple and small size keys, it lacked security. If the attacker were to figure out the key to an symmetric encrypted message, then they would easily be able to decipher it by using it.

This is why over the years the concept of larger key size and key diversity has been introduced. For example, DES made a huge impact on the world of symmetric encryption system. It was the foundation for the creation of AES and more secure symmetric encryption systems (Ketha, 2024).

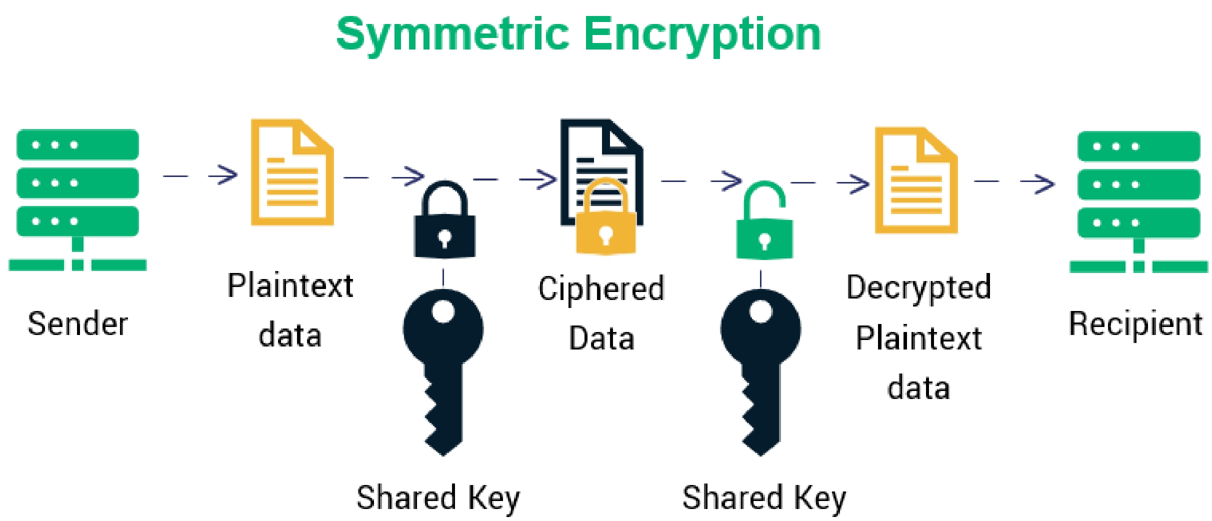

Figure 6: Symmetric encryption system (Thakkar, 2020)

2.2.2 Asymmetric encryption system

Asymmetric encryption system is a type of encryption system that uses two keys to encipher and decipher the message. The key that is used for encryption is called the public key while the key to decipher is called the private key (Ketha, 2024). This system is also called public key cryptosystem as the public eye is known to everyone and not only the sender and recipient. However, the private key is known to the recipient which is used to decipher the message. It is impossible to figure out the private key by just the public key. This is the core security concept of asymmetric encryption system (Shores, 2020). Some of the types of asymmetric encryption systems are RSA algorithm, differ-Hellman key exchange, digital signature algorithm (DSA) and Elliptic curve cryptography (ECC) (Ketha, 2024).

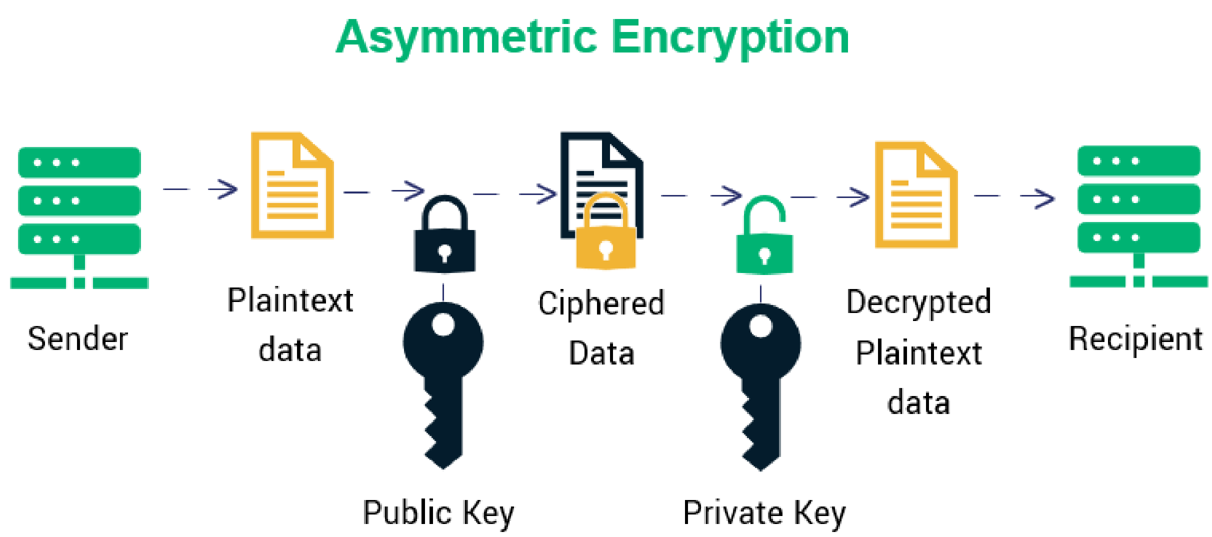

Figure 7: Asymmetric encryption system (Thakkar, 2020)

3 Background of the Selected Cryptographic Algorithm

3.1 Atbash Cipher

3.1.1 Introduction

Monoalphabetic substitution cipher are those ciphers that have only one cipher substitute for every letter. “Mono” refers to as one so one plain letter has one cipher substitute. Atbash Cipher is one of the examples that come under mono alphabetic ciphers.

3.1.2 Overview of Atbash Cipher

Atbash Cipher is one the many monoalphabetic substitution cipher that was created long time ago. It was made for the Hebrew alphabets in 600 BCE but can be used by any alphabets in the world. The key to this cipher is very simple as it just reverses the plaintext alphabet into the cipher alphabet. This means that the first letter of the alphabet is ciphered into the last letter in the alphabet, the second letter in the alphabet becomes the second to the last letter of the alphabet and so on (Yadav, 2025).

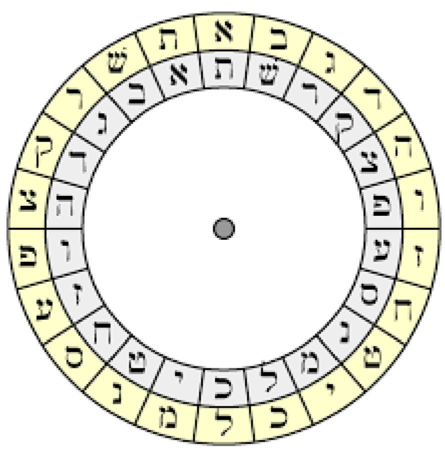

Figure 8: Atbash Cipher in Hebrew (Gondaliya, 2020)

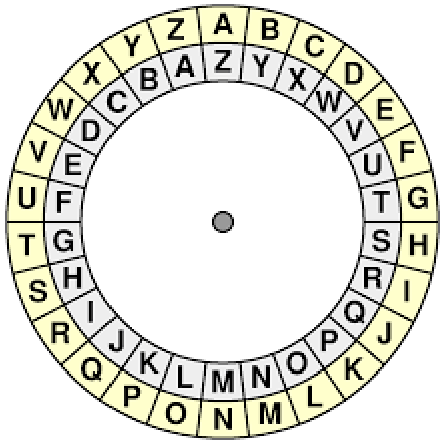

Figure 9: Atbash cipher in English Alphabet (Gondaliya, 2020)

The above figure shows that every letter's atbash cipher letter is the

The enciphering process of this cipher uses the following formula:

N = 25 – O

Where, O is short for old position number and N is short for new position number. This formula assumes a 26 letters alphabet where A=0…Z=25 (LetterToNumber, 2025).

3.1.3 Example:

An example of using Atbash Cipher to encrypt and decrypt a word is given below: Plaintext= "HACK"

Now, enciphering the plaintext in a table according to atbash cipher:

Table 1: Encryption table of Atbash Cipher

Plaintext letter	Old position number	N=25-O	New position number	Encrypted text
H	7	N= 25-7	18	S
A	0	N= 25-0	25	Z
C	2	N= 25-2	23	X
K	10	N=25-10	15	P

The encrypted text of the plaintext “HACK” is “SZXP”

Now, deciphering the encrypted text in a table according to atbash cipher:

Table 2: Decryption table of Atbash Cipher

Encrypted letters	New position number	O= 25-N	Old position number	Decrypted letters
S	18	O= 25-18	7	H
Z	25	O=25-15	0	A
X	23	O=25-23	2	C
P	15	O=25-15	10	K

3.2 Advantages and Disadvantages of Atbash Cipher

3.2.1 Advantages of Atbash Cipher

- Simple to implement: Atbash Cipher only requires the implementation of one key that is re positioning the letter with the corresponding position in the end of the alphabet. It is very simple to encipher this way (Cipher Utility, 2025).

- Self-inverse: The formula that is used to encipher is same as the formula used to decipher. This shows how the sender and receiver don’t have struggle to encipher and decipher (Cipher Utility, 2025).

- Historical significance: Many modern ciphers have been created yet; it is the classical ciphers that must be given the credit. Classical ciphers like atbash cipher helped to develop modern cryptographic tools (Cipher Utility, 2025).

- Implements on any alphabets: The atbash cipher can be used for any types of alphabet system (Yadav, 2025). This makes it available to everyone without having to modify anything.

- Quicker manual calculations: The calculations for atbash can be done manually, without requiring high tech computers. This makes the message reach the receiver quickly who will just have to do simple maths for deciphering (Cipher Utility, 2025).

3.2.2 Disadvantages of Atbash Cipher

- Easily decipherable: The atbash cipher was considered safe in the ancient times as many people weren’t as educated as today. Simple computational capabilities can decipher it now. This shows that it is no longer secure and should not be relied upon (Yadav, 2025).

- Lack of security: The atbash cipher is very easy to implement and this is why there is high chances of security risks. The attackers can decipher the message easily as it was enciphered (Yadav, 2025).

- Vulnerability to frequency analysis: The atbash ciphers replaces one letter with another in a fixed manner. So, attackers can use frequency analysis and see what letter is replaced by what letter especially when the cipher text is longer (Hassan & Dakingari, 2025).

- Weak compared to the modern ciphers: The modern ciphers have many complex mathematical and logical operations involved. They make the plaintext impossible to be deciphered by anyone but the receiver (Hassan & Dakingari, 2025). Atbash on the other hand, can be deciphered by anyone if they know the reverse alphabets.

- Lack of applicability in the modern world: Despite this cipher being popular in the ancient times, it would not be applicable today. There are stronger encryption mechanisms and attackers have many stronger decryption tools. It cannot secure messages like it used to.

4 Development of new cryptographic algorithm

4.1 Modifications for Atbash Cipher

4.1.1 Mathematical modifications in encryption

- Decimal number conversion to Binary: The position numbers in the original atbash is quite guessable. This is why in the new algorithm; there is the inclusion of converting those decimals to binary and doing operations with them. There are many frequent conversions of binary to decimal and decimal to binary.

- Integrating key decimal: The atbash cipher used an easy formula that could be exploited by the attackers so, in the new algorithm this formula (decimal+key decimal) MOD 32 has been used. Unlike the old formula, this formula requires the decimals of the key letters. The third party cannot use this formula unless they know the key. However, even if this formula is cracked the answer will still be encrypted as many other encryption mechanisms have been used.

- Decimals not greater or equal to 32: The positional values of the alphabets range from 0 to 25 in the original atbash. However, in the new algorithm there will be 32 positional values i.e. 0 to 31 where the following symbols will be added:

Table 3: Additional new characters

Symbols	Decimal positional value
!	26
@	27
#	28
$	29
%	30
^	31

The MOD 32 is also useful as the maximum value of the 5-digit value is 31. This is why additional characters were added to the new algorithm.

- Modified Atbash cipher formula: The old atbash formula was “New position= 25-Old position” that substituted the letters. In this modified formula instead of 25, 31 is used. The more letters there are the more complex the cipher will become.

4.1.2 Logical modifications in encryption

- Rotate left by one: Rotation towards the left by one means each bit in the 5-digit binary shift in the left direction one by one. It has been done in the encryption algorithm when the plain letters were converted into 5-digit binaries.

- NOT operation: NOT operation means that each of the binary bits have been converted into the opposite like 0 to 1 and 1 to 0. In the Jelly algorithm, the NOT operation is used right after the left rotation operation to make the binary digits more complex.

4.1.3 Mathematical modifications in decryption

The decryption method in original Atbash cipher only required the use of O= 25-N, where O, represents the original or old position number of the letter and “N” represents the new or current position number of the letter. This single formula was used to decrypt the atbash cipher. It may be easier to use but it also means that it can be very insecure. This why the following modifications have been made:

- Conversions from decimal to binary: The decimal values and the binary values have been converted into each other multiple times in the encryption algorithm so, it is also required in the decryption algorithm.

- Integrating key decimals: In the new algorithm, the corresponding key decimal is subtracted from each cipher text number, then computed by MOD 32.

- Modified Atbash cipher formula: The old atbash formula was “New_position= 25-Old_position” that substituted the letters. In this modified formula instead of 25, 31 is used to reverse the encryption atbash.

4.1.4 Logical modifications in decryption

- Rotate right by one: To reverse the left rotation that was done on the encryption process, right rotation must be done. It is the exact opposite of the left rotation where each bit is shifted towards the not left but right direction.

- Reverse NOT operation: The NOT operation done in the encryption process can only be reversed if it is done again on the decryption process.

4.2 Jelly Cipher algorithm

4.2.1 Encryption Algorithm

Table 4: Encryption Algorithm

Steps	Description
Step 1	Enter a plaintext and a key
Step 2	Convert the inputted plaintext into uppercase
Step 3	Convert the given key into uppercase and match it with the length of the plaintext
Step 4	Convert each of the plaintext letters into the decimal positional value i.e A=0, B=1…^=31
Step 5	Convert each letter's decimal into 5-bit binary
Step 6	Rotate left by one on each of the binary groups
Step 7	Apply bitwise NOT operation on the binary groups
Step 8	Convert the resulting binary into decimal
Step 9	Add the corresponding key decimal to each resulting decimal, then compute with MOD 32
Step 10	Apply modified atbash cipher formula by subtracting the old position number of the letter by 31 to get the new position number
Step 11	Convert the decimal result into the corresponding letter
Step 12	Combine all the letters to form ciphertext
Step 13	Display the ciphertext by changing to lowercase.

4.2.2 Decryption Algorithm

Table 5: Decryption Algorithm

Steps	Description
Step 1	Convert the given ciphertext into uppercase.
Step 2	Convert key to uppercase and repeat it to match the length of ciphertext.
Step 3	Apply modified atbash cipher formula by subtracting the new position number of the letter by 31 to get the old position number.
Step 4	Convert each of the ciphertext letters into the decimal positional value i.e. A=0, B=1…^=31.
Step 5	Subtract the corresponding key decimal from each positional value, then compute with MOD 32.
Step 6	Convert each decimal to 5-bit binary.
Step 7	Apply bitwise NOT operation on the binary groups to reverse encryption.
Step 8	Rotate right by one on each of the binary groups
Step 9	Convert each binary back to decimal.
Step 10	Convert the decimal result into the corresponding letter.
Step 11	Combine all letters to form the plaintext.
Step 12	Display the plaintext by changing to lowercase.

4.3 Flowchart of Jelly Cipher

4.3.1 Encryption Flowchart

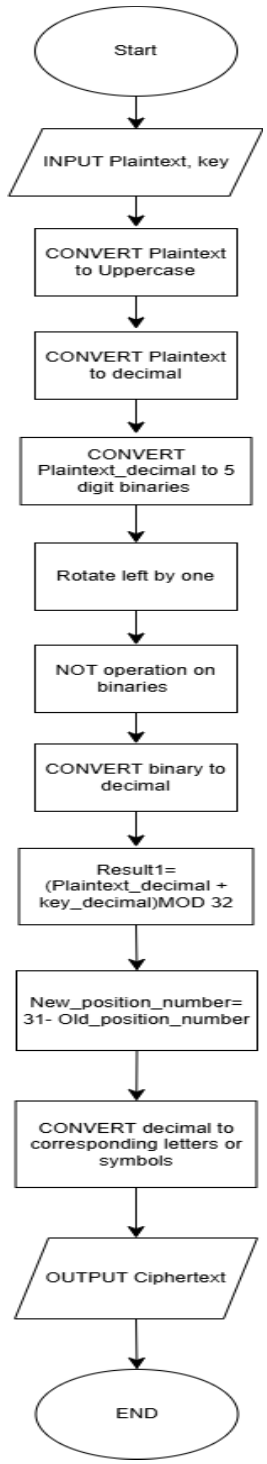

Figure 10: Encryption Flowchart

4.3.2 Decryption Flowchart

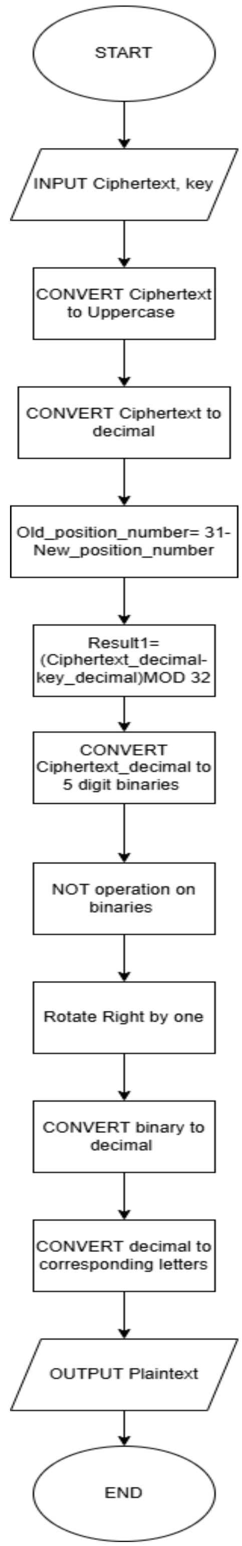

Figure 11: Decryption Flowchart

4.4 Example of Jelly Cipher

The plaintext and key are given below,

Plaintext= Dark

Key= Star

4.4.1 Encryption process

Step 1: Convert the plaintext into uppercase

Table 6: Lowercase to Uppercase

Lowercase	Uppercase
Dark	DARK

Step 2: Convert key to uppercase and match it with the length of the plaintext.

Table 7: Matching key length to plaintext

Lowercase	Uppercase
Star	STAR

DARK and STAR both have the same word length.

Step 3: Convert the plaintext letters into decimal positional values

Table 8: Plaintext letters to decimal positional values

Plaintext letters	Decimal positional values
D	3
A	0
R	17
K	10

Step 4: Convert each decimal to 5-digit binary value

Table 9: Conversion to Binary

Decimal positional values	Binary values
3	00011
0	00000
17	10001
10	01010

Step 5: Rotate left by one on each of the binary groups

Table 10: Rotate left by one

Binary values	Rotate left by one
00011	00110
00000	00000
10001	00011
01010	10100

Step 6: Apply bitwise NOT operation on the binary groups

Table 11: Bitwise NOT

Result after rotation	Bitwise NOT operation
00110	11001
00000	11111
00011	11100
10100	01011

Step 7: Convert the resulting binary into decimal

Table 12: Conversion to Decimal

Result after bitwise NOT	Decimal

11001	25
11111	31
11100	28
01011	11

Step 8: Add the corresponding key decimal to each resulting decimal, then compute with MOD 32.

Table 13: Mathematical operation

Decimal	Key letters	Key Decimal	(Decimal + key decimal) mod 32	Result
25	S	18	(25+18) MOD 32	11
31	T	19	(31+19) MOD 32	18
28	A	0	(28+0) MOD 32	28
11	R	17	(11+ 17) MOD 32	28

Step 9: Apply modified Atbash cipher formula

Table 14: Modified Atbash Cipher Forumla

Decimal	Letter	Modified Atbash cipher formula	Result decimal	Corresponding letter
11	L	N=31- 11	20	U
18	S	N= 31-18	13	N
28	#	N= 31-28	3	D

28	#	N= 31-28	3	D

Step 10: Combine all letters to form the ciphertext

Ciphertext = UNDD

Step 12: Display the Ciphertext

Undd

4.4.2 Decryption process

The Ciphertext and key are given below,

Ciphertext= Undd

Key= Star

Step 1: Convert the given ciphertext into uppercase

Table 15: Lowercase to Uppercase

Lowercase	Uppercase
Undd	UNDD

Step 2: Convert key to uppercase and repeat it to match the length of ciphertext.

Table 16: Matching key length to Ciphertext

Lowercase	Uppercase
Star	STAR

Step 3: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 17: Letters to decimals

Ciphertext letters	Decimal positional Values

U	20
N	13
D	3
D	3

Step 4: Decrypt the atbash cipher

Table 18: Decryption of atbash cipher

Letter	Modified Atbash formula	Decimal Result	Cipher letter
U	N= 31-20	11	L
N	N= 31-13	18	S
D	N= 31-3	28	#
D	N= 31-3	28	#

Step 5: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 19: Conversion to Decimal

Ciphertext letters	Decimal positional values
L	11
S	18
#	28
#	28

Step 6: Subtract the corresponding key decimal from each positional value, then compute with MOD 32.

Table 20: Mathematical Operation

Decimal	Key letters	Key Decimal	(Decimal-key decimal) mod 32	Result
11	S	18	(11-18) MOD 32	25
18	T	19	(18-19) MOD 32	31
28	A	0	(28-0) MOD 32	28
28	R	17	(28-17) MOD 32	11

Step 7: Convert each decimal to 5-bit binary.

Table 21: Conversion to binary

Decimal positional values	Binary values
25	11001
31	11111
28	11100
11	01011

Step 8: Apply bitwise NOT operation on the binary groups to reverse encryption.

Table 22: Bitwise NOT operation

Binary Values	Bitwise NOT operation
11001	00110
11111	00000
11100	00011

01011	10100

Step 9: Rotate right by one on each of the binary groups

Table 23: Rotate Right by one

Binary values after NOT	Rotate right by one
00110	00011
00000	00000
00011	10001
10100	01010

Step 10: Convert each binary back to decimal

Table 24: Binary to Decimal

Binary	Decimal value
00011	3
00000	0
10001	17
01010	10

Step 11: Convert the decimal result into the corresponding letter.

Table 25: Conversion to letters

Decimal Value	Corresponding letter
3	D
0	A
17	R
10	K

Step 12: Combine all letters to form the plaintext.

Plaintext= DARK

Step 12: Display the plaintext.

Dark

5 Test Cases

5.1 Test 1

Test Objective: To test the encryption and decryption of the algorithm by providing equal plaintext and key length words.

The plaintext and key are given below,

Plaintext= Photo

Key= Mouse

5.1.1 Encryption process for Test 1

Step 1: Convert the plaintext into uppercase

Table 26: Lowercase to Uppercase

Lowercase	Uppercase
Photo	PHOTO

Step 2: Convert key to uppercase and match it with the length of the plaintext.

Table 27: Matching key length to plaintext

Lowercase	Uppercase
Mouse	MOUSE

PHOTO and MOUSE both have the same word length.

Step 3: Convert the plaintext letters into decimal positional values

Table 28: Plaintext letters to decimal positional values

Plaintext letters	Decimal positional values
P	15
H	7
O	14

T	19
O	14

Step 4: Convert each decimal to 5-digit binary value

Table 29: Conversion to Binary

Decimal positional values	Binary values
15	01111
7	00111
14	01110
19	10011
14	01110

Step 5: Rotate left by one on each of the binary groups

Table 30: Rotate Left by one

Binary values	Rotate Left by one
01111	11110
00111	01110
01110	11100
10011	00111
01110	11100

Step 6: Apply bitwise NOT operation on the binary groups

Table 31: Bitwise NOT

Result after rotation	Bitwise NOT operation
11110	00001
01110	10001
11100	00011

00111	11000
11100	00011

Step 7: Convert the resulting binary into decimal

Table 32: Conversion to Decimal

Result after bitwise NOT	Decimal
00001	1
10001	17
00011	3
11000	24
00011	3

Step 8: Add the corresponding key decimal to each resulting decimal, then compute with MOD 32.

Table 33: Mathematical operation

Decimal	Key letters	Key Decimal	(Decimal + key decimal) mod 32	Result
1	M	12	(1+12) MOD 32	13
17	O	14	(17+14) MOD 32	31
3	U	20	(3+20) MOD 32	23
24	S	18	(24+18) MOD 32	10
3	E	4	(3+4) MOD 32	7

Step 9: Apply modified Atbash cipher formula

Table 34: Modified Atbash cipher formula

Decimal	Letter	Modified Atbash cipher formula	Result decimal	Corresponding letter
13	N	N= 31-13	18	S
31	^	N= 31-31	0	A
23	X	N=31-23	8	I
10	K	N=31-10	21	V
7	H	N=31-7	24	Y

Step 10: Combine all letters to form the ciphertext

Ciphertext = SAIVY

Step 11: Display the Ciphertext

Saivy

5.1.2 Decryption process for Test 1

The Ciphertext and key are given below,

Ciphertext= Saivy

Key= Mouse

Step 1: Convert the given ciphertext into uppercase

Table 35: Lowercase to Uppercase

Lowercase	Uppercase
Saivy	SAIVY

Step 2: Convert key to uppercase and repeat it to match the length of ciphertext.

Table 36: Matching key length to Ciphertext

Lowercase	Uppercase
Mouse	MOUSE

Step 3: Apply modified atbash cipher formula by subtracting the new position number of the letter by 31 to get the old position number

Table 37: Modified Atbash Cipher formula

Letter	Decimal	Modified Atbash cipher formula	Result decimal	Corresponding letter
S	18	N= 31-18	13	N
A	0	N= 31-0	31	^
I	8	N=31-8	23	X
V	21	N=31-21	10	K
Y	24	N=31-24	7	H

Step 4: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 38: Conversion to Decimal

Ciphertext letters	Decimal positional values
N	13
^	31
X	23
K	10
H	7

Step 5: Subtract the corresponding key decimal from each positional value, then compute with MOD 32.

Table 39: Mathematical Operation

Decimal	Key letters	Key Decimal	(Decimal-key decimal) mod 32	Result
13	M	12	(13-12) MOD 32	1
31	O	14	(31-14) MOD 32	17
23	U	20	(23-20) MOD 32	3
10	S	18	(10-18) MOD 32	24
7	E	4	(7-4) MOD 32	3

Step 6: Convert each decimal to 5-bit binary.

Table 40: Conversion to binary

Decimal positional values	Binary values
1	00001
17	10001
3	00011
24	11000
3	00011

Step 7: Apply bitwise NOT operation on the binary groups to reverse encryption.

Table 41: Bitwise NOT operation

Binary Values	Bitwise NOT operation
00001	11110
10001	01110
00011	11100
11000	00111
00011	11100

Step 8: Rotate right by one on each of the binary groups

Table 42: Rotate right by one

Binary values after NOT	Rotate right by one
11110	01111
01110	00111
11100	01110
00111	10011
11100	01110

Step 9: Convert each binary back to decimal

Table 43: Binary to Decimal

Binary	Decimal value
01111	15
00111	7
01110	14
10011	19
01110	14

Step 10: Convert the decimal result into the corresponding letter.

Table 44: Conversion to letters

Decimal Value	Corresponding letter
15	P
7	H
14	O
19	T
14	O

Step 11: Combine all letters to form the plaintext.

Plaintext= PHOTO

Step 12: Display the plaintext.

PHOTO

Therefore, the Test 1 has been proven to be successful.

5.2 Test 2

Test Objective: To test the encryption and decryption of the algorithm by longer length plaintext and shorter length key word.

The plaintext and key are given below,

Plaintext= People

Key= Demon

5.2.1 Encryption process for Test 2

Step 1: Convert the plaintext into uppercase

Table 45: Lowercase to Uppercase

Lowercase	Uppercase
People	PEOPLE

Step 2: Convert key to uppercase and match it with the length of the plaintext.

Table 46: Matching key length to plaintext

Lowercase	Uppercase
Demon	DEMOND

PEOPLE and DEMON do not have the same word length so, the key word has repeated the first letter in the end. This produces DEMOND key word.

Step 3: Convert the plaintext letters into decimal positional values

Table 47: Plaintext letters to decimal positional values

Plaintext letters	Decimal positional values
P	15
E	4
O	14
P	15
L	11
E	4

Step 4: Convert each decimal to 5-digit binary value

Table 48: Conversion to Binary

Decimal positional values	Binary values
15	01111

4	00100
14	01110
15	01111
11	01011
4	00100

Step 5: Rotate left by one on each of the binary groups

Table 49: Rotate left by one

Binary values	Rotate left by one
01111	11110
00100	01000
01110	11100
01111	11110
01011	10110
00100	01000

Step 6: Apply bitwise NOT operation on the binary groups

Table 50: Bitwise NOT

Result after rotation	Bitwise NOT operation
11110	00001
01000	10111
11100	00011

11110	00001
10110	01001
01000	10111

Step 7: Convert the resulting binary into decimal

Table 51: Conversion to Decimal

Result after bitwise NOT	Decimal
00001	1
10111	23
00011	3
00001	1
01001	9
10111	23

Step 8: Add the corresponding key decimal to each resulting decimal, then compute with MOD 32.

Table 52: Mathematical operation

Decimal	Key letters	Key Decimal	(Decimal + key decimal) mod 32	Result
1	D	3	(1+3) MOD 32	4
23	E	4	(23+4) MOD 32	27

3	M	12	(3+12) MOD 32	15
1	O	14	(1+14) MOD 32	15
9	N	13	(9+13) MOD 32	22
23	D	3	(23+3) MOD 32	26

Step 9: Apply modified Atbash cipher formula

Table 53: Modified Atbash cipher formula

Decimal	Letter	Modified Atbash cipher formula	Result decimal	Corresponding letter
4	E	N= 31-4	27	@
27	@	N= 31-27	4	E
15	P	N=31-15	16	Q
15	P	N=31-15	16	Q
22	W	N=31-22	9	J
26	!	N=31-26	5	F

Step 10: Combine all letters to form the ciphertext

Ciphertext = @EQQJF

Step 11: Display the Ciphertext

@Eqqjf

5.2.2 Decryption process for Test 2

The Ciphertext and key are given below,

Ciphertext= @Eqqjf

Key= Demon

Step 1: Convert the given ciphertext into uppercase

Table 54: Lowercase to Uppercase

Lowercase	Uppercase
@Eqqjf	@EQQJF

Step 2: Convert key to uppercase and repeat it to match the length of ciphertext.

Table 55: Matching key length to Ciphertext

Lowercase	Uppercase
Demon	DEMOND

@EQQJF and DEMON do not have the same word length so, the key word has repeated the first letter in the end. This produces DEMOND key word

Step 3: Apply modified atbash cipher formula by subtracting the new position number of the letter by 31 to get the old position number

Table 56: Modified Atbash Cipher formula

Letter	Decimal	Modified Atbash cipher formula	Result decimal	Corresponding letter
@	27	N= 31-27	4	E
E	4	N= 31-4	27	@

Q	16	N=31-16	15	P
Q	16	N=31-16	15	P
J	13	N=31-9	22	W
F	5	N=31-5	26	!

Step 4: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 57: Conversion to Decimal

Ciphertext letters	Decimal positional values
E	4
@	27
P	15
P	15
W	22
!	26

Step 5: Subtract the corresponding key decimal from each positional value, then compute with MOD 32.

Table 58: Mathematical Operation

Decimal	Key letters	Key Decimal	(Decimal-key decimal) mod 32	Result
4	D	3	(4-3) MOD 32	1

27	E	4	(27-4) MOD 32	23
15	M	12	(15-12) MOD 32	3
15	O	14	(15-14) MOD 32	1
22	N	13	(22-13) MOD 32	9
26	D	3	(26-3) MOD 32	23

Step 6: Convert each decimal to 5-bit binary.

Table 59: Conversion to binary

Decimal positional values	Binary values
1	00001
23	10111
3	00011
1	00001
9	01001
23	10111

Step 7: Apply bitwise NOT operation on the binary groups to reverse encryption.

Table 60: Bitwise NOT operation

Binary Values	Bitwise NOT operation
00001	11110
10111	01000
00011	11100

00001	11110
01001	10110
10111	01000

Step 8: Rotate right by one on each of the binary groups

Table 61: Rotate right by one

Binary values after NOT	Rotate right by one
11110	01111
01000	00100
11100	01110
11110	01111
10110	01011
01000	00100

Step 9: Convert each binary back to decimal

Table 62: Binary to Decimal

Binary	Decimal value
01111	15
00100	4
01110	14
01111	15
01011	11

00100	4

Step 10: Convert the decimal result into the corresponding letter.

Table 63: Conversion to letters

Decimal Value	Corresponding letter
15	P
4	E
14	O
15	P
11	L
4	E

Step 11: Combine all letters to form the plaintext.

Plaintext= PEOPLE

Step 12: Display the plaintext.

People

Therefore, the Test 2 has been proven to be successful.

5.3 Test 3

Test Objective: To test the encryption and decryption of the algorithm by providing shorter length plaintext and larger length key word.

The plaintext and key are given below,

Plaintext= Circle

Key= Rectangle

5.3.1 Encryption process for Test 3

Step 1: Convert the plaintext into uppercase

Table 64: Lowercase to Uppercase

Lowercase	Uppercase
Circle	CIRCLE

Step 2: Convert key to uppercase and match it with the length of the plaintext.

Table 65: Matching key length to plaintext

Lowercase	Uppercase
Rectangle	RECTAN

CIRCLE and RECTAN do not have the same word length so, the key word is reduced to only six letters to match the plaintext length. This produces RECTAN key word.

Step 3: Convert the plaintext letters into decimal positional values

Table 66: Plaintext letters to decimal positional values

Plaintext letters	Decimal positional values
C	2
I	8
R	17
C	2
L	11
E	4

Step 4: Convert each decimal to 5-digit binary value

Table 67: Conversion to Binary

Decimal positional values	Binary values
2	00010
8	01000
17	10001
2	00010
11	01011
4	00100

Step 5: Rotate left by one on each of the binary groups

Table 68: Rotate left by one

Binary values	Rotate left by one
00010	00100
01000	10000
10001	00011
00010	00100
01011	10110
00100	01000

Step 6: Apply bitwise NOT operation on the binary groups

Table 69: Bitwise NOT

Result after rotation	Bitwise NOT operation
00100	11011
10000	01111
00011	11100
00100	11011
10110	01001
01000	10111

Step 7: Convert the resulting binary into decimal

Table 70: Conversion to Decimal

Result after bitwise NOT	Decimal
11011	27
01111	15
11100	28
11011	27
01001	9
10111	23

Step 8: Add the corresponding key decimal to each resulting decimal, then compute with MOD 32.

Table 71: Mathematical operation

Decimal	Key letters	Key Decimal	(Decimal + key decimal) mod 32	Result
27	R	17	(27+17) MOD 32	12
15	E	4	(15+4) MOD 32	19
28	C	2	(28+2) MOD 32	30
27	T	19	(27+19) MOD 32	14
9	A	0	(9+0) MOD 32	9
23	N	13	(23+13) MOD 32	4

Step 9: Apply modified Atbash cipher formula

Table 72: Modified Atbash cipher formula

Decimal	Letter	Modified Atbash cipher formula	Result decimal	Corresponding letter
12	M	N= 31-12	19	T
19	T	N= 31-19	12	M
30	%	N=31-30	1	B
14	O	N=31-14	17	R

9	J	N=31-9	22	W
4	E	N=31-4	27	@

Step 10: Combine all letters to form the ciphertext

Ciphertext = TMBRW@

Step 11: Display the Ciphertext

Tmbrw@

5.3.2 Decryption process for Test 3

The Ciphertext and key are given below,

Ciphertext= Tmbrw@

Key= Rectangle

Step 1: Convert the given ciphertext into uppercase

Table 73: Lowercase to Uppercase

Lowercase	Uppercase
Tmbrw@	TMBRW@

Step 2: Convert key to uppercase and repeat it to match the length of ciphertext.

Table 74: Matching key length to Ciphertext

Lowercase	Uppercase
Rectangle	RECTAN

TMBRW@ and RECTAN do not have the same word length so, the key word is reduced to only six letters to match the ciphertext length. This produces RECTAN key word.

Step 3: Apply modified atbash cipher formula by subtracting the new position number of the letter by 31 to get the old position number

Table 75: Modified Atbash Cipher formula

Letter	Decimal	Modified Atbash cipher formula	Result decimal	Corresponding letter
T	19	N= 31-19	12	M
M	12	N= 31-12	19	T
B	1	N=31-1	30	%
R	17	N=31-17	14	O
W	22	N=31-22	9	J
@	27	N=31-27	4	E

Step 4: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 76: Conversion to Decimal

Ciphertext letters	Decimal positional values
M	12
T	19
%	30
O	14
J	9
E	4

Step 5: Subtract the corresponding key decimal from each positional value, then compute with MOD 32.

Table 77: Mathematical Operation

Decimal	Key letters	Key Decimal	(Decimal-key decimal) mod 32	Result
12	R	17	(12-17) MOD 32	27
19	E	4	(19-4) MOD 32	15
30	C	2	(30-2) MOD 32	28
14	T	19	(14-19) MOD 32	27
9	A	0	(9-0) MOD 32	9
4	N	13	(4-13) MOD 32	23

Step 6: Convert each decimal to 5-bit binary.

Table 78: Conversion to binary

Decimal positional values	Binary values
27	11011
15	01111
28	11100
27	11011
9	01001
23	10111

Step 7: Apply bitwise NOT operation on the binary groups to reverse encryption.

Table 79: Bitwise NOT operation

Binary Values	Bitwise NOT operation
11011	00100
01111	10000
11100	00011
11011	00100
01001	10110
10111	01000

Step 8: Rotate right by one on each of the binary groups

Table 80: Rotate right by one

Binary values after NOT	Rotate right by one
00100	00010
10000	01000
00011	10001
00100	00010
10110	01011
01000	00100

Step 9: Convert each binary back to decimal

Table 81: Binary to Decimal

Binary	Decimal value

00010	2
01000	8
10001	17
00010	2
01011	11
00100	4

Step 10: Convert the decimal result into the corresponding letter.

Table 82: Conversion to letters

Decimal Value	Corresponding letter
2	C
8	I
17	R
2	C
11	L
4	E

Step 11: Combine all letters to form the plaintext.

Plaintext= CIRCLE

Step 12: Display the plaintext.

Circle

Therefore, the Test 3 has been proven to be successful.

5.4 Test 4

Test Objective: To test the encryption and decryption of the algorithm by providing symbol in the plaintext.

The plaintext and key are given below,

Plaintext= Bowling#

Key= Cat

5.4.1 Encryption process for Test 4

Step 1: Convert the plaintext into uppercase

Table 83: Lowercase to Uppercase

Lowercase	Uppercase
Bowling	BOWLING#

Step 2: Convert key to uppercase and match it with the length of the plaintext.

Table 84: Matching key length to plaintext

Lowercase	Uppercase
Cat	CATCATCA

BOWLING# and CAT do not have the same word length so, the key word is repeated till the length matches. This produces CATCATCA key word.

Step 3: Convert the plaintext letters into decimal positional values

Table 85: Plaintext letters to decimal positional values

Plaintext letters	Decimal positional values
B	1

O	14
W	22
L	11
I	8
N	13
G	6
#	28

Step 4: Convert each decimal to 5-digit binary value

Table 86: Conversion to Binary

Decimal positional values	Binary values
1	00001
14	01110
22	10110
11	01011
8	01000
13	01101
6	00110
28	11100

Step 5: Rotate left by one on the binary values

Table 87: Rotate left by one

Binary values	Rotate left by one
00001	00010
01110	11100
10110	01101
01011	10110
01000	10000
01101	11010
00110	01100
11100	11001

Step 6: Apply bitwise NOT operation on the binary groups

Table 88: Bitwise NOT

Result after rotation	Bitwise NOT operation
00010	11101
11100	00011
01101	10010
10110	01001
10000	01111
11010	00101
01100	10011
11001	00110

Step 7: Convert the resulting binary into decimal

Table 89: Conversion to Decimal

Result after bitwise NOT	Decimal
11101	29
00011	3
10010	18
01001	9
01111	15
00101	5
10011	19
00110	6

Step 8: Add the corresponding key decimal to each resulting decimal, then compute with MOD 32.

Table 90: Mathematical operation

Decimal	Key letters	Key Decimal	(Decimal + key decimal) mod 32	Result
29	C	2	(29+2) MOD 32	31
3	A	0	(3+0) MOD 32	3

18	T	19	(18+19) MOD 32	5
9	C	2	(9+2) MOD 32	11
15	A	0	(15+0) MOD 32	15
5	T	19	(5+19) MOD 32	24
19	C	2	(19+2) MOD 32	21
6	A	0	(6+0) MOD 32	6

Step 9: Apply modified Atbash cipher formula

Table 91: Modified Atbash cipher formula

Decimal	Letter	Modified Atbash cipher formula	Result decimal	Corresponding letter
31	^	N= 31-31	0	A
3	D	N= 31-3	28	#
5	F	N=31-5	26	!
11	L	N=31-11	20	U
15	P	N=31-15	16	Q
24	Y	N=31-24	7	H
21	V	N=31-21	10	K
6	G	N= 31-6	25	Z

Step 10: Combine all letters to form the ciphertext

Ciphertext = A#!UQHKZ

Step 11: Display the Ciphertext

a#!uqhkz

5.4.2 Decryption process for Test 4

The Ciphertext and key are given below,

Ciphertext= a#!uqhkz

Key= Cat

Step 1: Convert the given ciphertext into uppercase

Table 92: Lowercase to Uppercase

Lowercase	Uppercase
a#!uqhkz	A#!UQHKZ

Step 2: Convert key to uppercase and repeat it to match the length of ciphertext.

Table 93: Matching key length to Ciphertext

Lowercase	Uppercase
Cat	CATCATCA

A#!UQHKZ and CAT do not have the same word length so, the key word is repeated till the length matches. This produces CATCATCA key word.

Step 3: Apply modified atbash cipher formula by subtracting the new position number of the letter by 31 to get the old position number

Table 94: Modified Atbash Cipher formula

Letter	Decimal	Modified Atbash cipher formula	Result decimal	Corresponding letter
A	0	N= 31-0	31	^
#	28	N= 31-28	3	D
!	26	N=31-26	5	F
U	20	N=31-20	11	L
Q	16	N=31-16	15	P
H	7	N=31-7	24	Y
K	10	N=31-10	21	V
Z	25	N=31-25	6	G

Step 4: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 95: Conversion to Decimal

Ciphertext letters	Decimal positional values
^	31
D	3
F	5
L	11
P	15
Y	24

V	21
G	6

Step 5: Subtract the corresponding key decimal from each positional value, then compute with MOD 32.

Table 96: Mathematical Operation

Decimal	Key letters	Key Decimal	(Decimal-key decimal) mod 32	Result
31	C	2	(31-2) MOD 32	29
3	A	0	(3-0) MOD 32	3
5	T	19	(5-19) MOD 32	18
11	C	2	(11-2) MOD 32	9
15	A	0	(15-0) MOD 32	15
24	T	19	(24-19) MOD 32	5
21	C	2	(21-2) MOD 32	19
6	A	0	(6-0) MOD 32	6

Step 6: Convert each decimal to 5-bit binary.

Table 97: Conversion to binary

Decimal positional values	Binary values
29	11101
3	00011

18	10010
9	01001
15	01111
5	00101
19	10011
6	00110

Step 7: Apply bitwise NOT operation on the binary groups to reverse encryption.

Table 98: Bitwise NOT operation

Binary Values	Bitwise NOT operation
11101	00010
00011	11100
10010	01101
01001	10110
01111	10000
00101	11010
10011	01100
00110	11001

Step 8: Rotate right by one on each of the binary groups

Table 99: Rotate right by one

Binary values after NOT	Rotate right by one

00010	00001
11100	01110
01101	10110
10110	01011
10000	01000
11010	01101
01100	00110
11001	11100

Step 9: Convert each binary back to decimal

Table 100: Binary to Decimal

Binary	Decimal value
00001	1
01110	14
10110	22
01011	11
01000	8
01101	13
00110	6
11100	28

Step 10: Convert the decimal result into the corresponding letter.

Table 101: Conversion to letters

Decimal Value	Corresponding letter
1	B
14	O
22	W
11	L
8	I
13	N
6	G
28	#

Step 11: Combine all letters to form the plaintext.

Plaintext= BOWLING#

Step 12: Display the plaintext.

Bowling#

Therefore, the Test 4 has been proven to be successful.

5.5 Test 5

Test Objective: To test the encryption and decryption of the algorithm by providing a symbol in the keyword

The plaintext and key are given below,

Plaintext= Notebook

Key= Pencil@

5.5.1 Encryption process for Test 5

Step 1: Convert the plaintext into uppercase

Table 102: Lowercase to Uppercase

Lowercase	Uppercase
Notebook	NOTEBOOK

Step 2: Convert key to uppercase and match it with the length of the plaintext.

Table 103: Matching key length to plaintext

Lowercase	Uppercase
Pencil@	PENCIL@P

NOTEBOOK and PENCIL@ do not have the same word length so, the key word is repeated till the length matches. This produces PENCIL@P key word.

Step 3: Convert the plaintext letters into decimal positional values

Table 104: Plaintext letters to decimal positional values

Plaintext letters	Decimal positional values
N	13
O	14
T	19
E	4
B	1

O	14
O	14
K	10

Step 4: Convert each decimal to 5-digit binary value

Table 105: Conversion to Binary

Decimal positional values	Binary values
13	01101
14	01110
19	10011
4	00100
1	00001
14	01110
14	01110
10	01010

Step 5: Rotate left by one on the binary values

Table 106: Rotate left by one

Binary values	Rotate left by one
01101	11010
01110	11100
10011	00111

00100	01000
00001	00010
01110	11100
01110	11100
01010	10100

Step 6: Apply bitwise NOT operation on the binary groups

Table 107: Bitwise NOT

Result after rotation	Bitwise NOT operation
10110	01001
00111	11000
11001	00110
00010	11101
10000	01111
00111	11000
00111	11000
00101	11010

Step 7: Convert the resulting binary into decimal

Table 108: Conversion to Decimal

Result after bitwise NOT	Decimal
01001	9

11000	24
00110	6
11101	29
01111	15
11000	24
11000	24
11010	26

Step 8: Add the corresponding key decimal to each resulting decimal, then compute with MOD 32.

Table 109: Mathematical operation

Decimal	Key letters	Key Decimal	(Decimal + key decimal) mod 32	Result
9	P	15	(9+15) MOD 32	24
24	E	4	(24+4) MOD 32	28
6	N	13	(6+13) MOD 32	19
29	C	2	(29+2) MOD 32	31
15	I	8	(15+8) MOD 32	23

24	L	11	(24+11) MOD 32	3
24	@	27	(24+27) MOD 32	19
26	P	15	(26+15) MOD 32	9

Step 9: Apply modified Atbash cipher formula

Table 110: Modified Atbash cipher formula

Decimal	Letter	Modified Atbash cipher formula	Result decimal	Corresponding letter
24	Y	N= 31-24	7	H
28	#	N= 31-28	3	D
19	T	N=31-19	12	M
31	^	N=31-31	0	A
23	X	N=31-23	8	I
3	D	N=31-3	28	#
19	T	N=31-19	12	M
9	J	N=31-9	22	W

Step 10: Combine all letters to form the ciphertext

Ciphertext = HDMAI#MW

Step 11: Display the Ciphertext

Hdmai#mw

5.5.2 Decryption process for Test 5

The Ciphertext and key are given below,

Ciphertext= Hdmai#mw

Key= Pencil

Step 1: Convert the given ciphertext into uppercase

Table 111: Lowercase to Uppercase

Lowercase	Uppercase
Hdmai#mw	HDMAI#MW

Step 2: Convert key to uppercase and repeat it to match the length of ciphertext.

Table 112: Matching key length to Ciphertext

Lowercase	Uppercase
Pencil	PENCIL@P

HDMAI#YB and PENCIL@ do not have the same word length so, the key word is repeated till the length matches. This produces PENCIL@P key word.

Step 3: Apply modified atbash cipher formula by subtracting the new position number of the letter by 31 to get the old position number

Table 113: Modified Atbash Cipher formula

Letter	Decimal	Modified Atbash cipher formula	Result decimal	Corresponding letter
H	7	N= 31-7	24	Y

D	3	N= 31-3	28	#
M	12	N=31-12	19	T
A	0	N=31-0	31	^
I	8	N=31-8	23	X
#	28	N=31-28	3	D
M	12	N=31-12	19	T
W	22	N=31-22	9	J

Step 4: Convert each of the ciphertext letters into the decimal positional value i.e A=0, B=1…Z=25

Table 114: Conversion to Decimal

Ciphertext letters	Decimal positional values
Y	24
#	28
T	19
^	31
X	23
D	3
T	19
J	9

Step 5: Subtract the corresponding key decimal from each positional value, then compute with MOD 32.

Table 115: Mathematical Operation

Decimal	Key letters	Key Decimal	(Decimal-key decimal) mod 32	Result
24	P	15	(24-15) MOD 32	9
28	E	4	(28-4) MOD 32	24
19	N	13	(19-13) MOD 32	6
31	C	2	(31-2) MOD 32	29
23	I	8	(23-8) MOD 32	15
3	L	11	(3-11) MOD 32	24
19	@	27	(19-27) MOD 32	24
9	P	15	(9-15) MOD 32	26

Step 6: Convert each decimal to 5-bit binary.

Table 116: Conversion to binary

Decimal positional values	Binary values
9	01001
24	11000
6	00110
29	11101
15	01111
24	11000
24	11000

26	11010

Step 7: Apply bitwise NOT operation on the binary groups to reverse encryption.

Table 117: Bitwise NOT operation

Binary Values	Bitwise NOT operation
01001	10110
11000	00111
00110	11001
11101	00010
01111	10000
11000	00111
11000	00111
11010	00101

Step 8: Rotate right by one on each of the binary groups

Table 118: Rotate right by one

Binary values after NOT	Rotate right by one
10110	01101
00111	01110
11001	10011
00010	00100
10000	00001

00111	01110
00111	01110
00101	01010

Step 9: Convert each binary back to decimal

Table 119: Binary to Decimal

Binary	Decimal value
01101	13
01110	14
10011	19
00100	4
00001	1
01110	14
01110	14
01010	10

Step 10: Convert the decimal result into the corresponding letter.

Table 120: Conversion to letters

Decimal Value	Corresponding letter
13	N
14	O
19	T

4	E
1	B
14	O
14	O
10	K

Step 11: Combine all letters to form the plaintext.

Plaintext= NOTEBOOK

Step 12: Display the plaintext.

Notebook

Therefore, the Test 5 has been proven to be successful.

6 Critical Evaluation of the new cryptographic algorithm

6.1 Strengths

- Complex mathematical operations: The Jelly cipher includes complex mathematical operations to make the algorithm more secure and different than the atbash cipher.

- Addition of symbols: The character set used for encryption and decryption has the addition of symbols like !, @, #, $, and ^. These symbols make the cipher look unreadable and increases the number of characters to 32.

- Usage of key word: There has been the usage of key word for every enciphering process. The keyword can only be known to the sender and the receiver. The receiver can only decrypt the word or message by using the key word. A plus point is that the keyword can be decreased and increased according to the plain text so this brings more security.

- Secure than atbash cipher: This Jelly cryptographic algorithm is much secure than the atbash cipher as there are multiple steps involved that are logical and mathematical operations.

- Polyalphabetic substitution: The substitution of every letter in a word is not substituted it to the fixed letter. Instead, they are replaced with the letter that is result of multiple logical and mathematical operations.

6.2 Weaknesses

- Dependence on keyword: The Jelly cryptographic algorithm is heavily dependent on a single key word. If the keyword is known to a third-party member, there is possibility of the message to be decrypted by them. The length of the plaintext is equal to the length of the ciphertext so the attackers can decrease or increase the key word length and try to decrypt the information.

- Predictable positional numbers: The positional numbers of every letter in the new character set can be easily guessed by the third-party member. The A to ^ character set has 0 to 31 positional numbers which are sequential.

- Short bit-length: The bit-length use in the Jelly algorithm is only 5 bits. It is short and can be quickly changed into decimal and vice versa. The short bit-length also quickens the process of NOT operation and left shift.

- Limited characters: Despite the addition of the symbols, the character set comprises of only 32 characters i.e. A to ^. This is limited and can be cracked by the attackers easily.

- Unsuitable for modern cryptography world: The modern cryptography world has many complex algorithms like AES, DES, and many more. These algorithms are far more complex and have sophisticated mathematical and logical operations.

- Brute force attack: An attacker may brute force by using multiple keys for decryption. This cipher already is small so the attackers may quickly decipher the message.

6.3 Application Area

- Academic area: The Jelly cipher can be used in academic sector to learn how ciphers can be created. It creates a foundation for learners to understand what basic steps are required for creating a new cryptographic algorithm. There is usage of modulus, NOT, left shift, right shift and bitwise. These are some of the concepts that learners must comprehend before moving forward to cryptography in depth.

- Sensor nodes: There are some IOT systems that can use this cryptography to secure their data temporarily. The devices that have limited CPU power and small memory storage like sensor nodes. They only sense the physical data that is small and repetitive, process it and send it to other devices. For example, weather sensors, temperature sensor and so on (Vishal , et al., 2020).

- RFID tags: RFID tags have limited hardware and no battery. They store small amounts of information, and it is used for identification and tracking purposes. They use radio waves to send information. The jelly cipher can hide the basic authentication for this system and hide the data (Vishal , et al., 2020).

- Non-critical communication: When non-critical information is being shared between individuals that have no severe consequences then the Jelly cipher can be used. The communication is non-critical, so it is unlikely the high-level attackers take interest in it.

7 Conclusion

In conclusion, cryptography is a vast topic with multiple subjects like symmetric, asymmetric, algorithms, keys and many more. Over the years, cryptography has grown stronger and secure to be used in critical systems. There has been creation of AES, DES and other sophisticated cryptographic algorithms. The credit to their creation goes to classic ciphers like Caesar cipher or atbash cipher.

This report was focused on the modifications that could be done to atbash cipher by including mathematical and logical operations. This bought upon the creation of the new cryptographic algorithm called the Jelly cipher. It provides more security as it uses an improved character set, uses modulus, Bitwise, NOT operation, left rotation, right rotation and other operations.

Despite the modifications there could be seen some weaknesses like any other ciphers that have existed till today. The various application areas ranging for academic areas to some simple IOT systems shows the efficient usage of Jelly Cipher. Therefore, any cryptographic algorithm can be always improvised to become increasingly secure.

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9 Appendix A

In the west, they started to make progress in cryptography during the renaissance. The ciphers started to be more sophisticated and lead to the development of polyalphabetic Vigenère cipher (Sidhu, 2023). It was created by Blaise de Vigenère who was took assistance from Leon Battista Alberti's cipher disk which allowed letters to be replaced by different letters at different times (Pal, et al., 2020). The Vigenère cipher is the modified version of Caesar cipher as it uses different Caesar shifts for each letter (Sidhu, 2023).

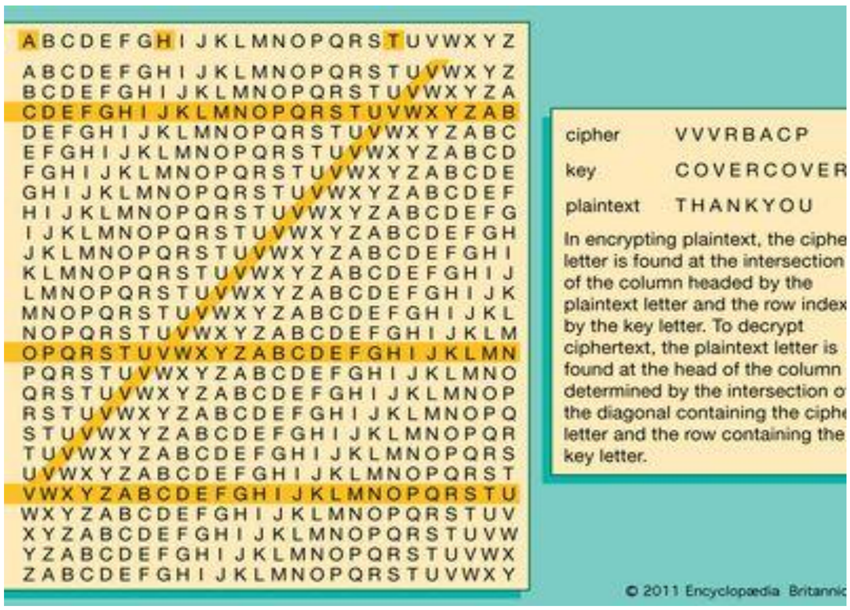

Figure 12: Vigenère Cipher (Simmons, 2025)